ALGEBRA 1 · PERSONALIZED PRACTICE

Math Practice

2026-11-13 · 20 questions

I CAN USE INPUT-OUTPUT RULES TO REPRESENT RELATIONSHIPS.

1. I can use input-output rules to represent relationships.

A machine adds 4 to an input, then squares the result. Explain why $(x + 4)^2$ matches but $x^2 + 4$ does not.

2. I can use input-output rules to represent relationships.

Input	-1	0	2
Output	1	3	7

Describe an operation rule that works for every pair.

3. I can use input-output rules to represent relationships.

A function machine triples the input and subtracts 2. What output comes from input 5?

4. I can use input-output rules to represent relationships.

Write a rule.

Inputs 0,1,2 give outputs 4,7,10.

I CAN DISTINGUISH FUNCTIONS FROM RELATIONSHIPS THAT ARE NOT FUNCTIONS.

5. I can distinguish functions from relationships that are not functions.

Function or not?

$(0, 1), (2, 3), (0, 4)$

6. I can distinguish functions from relationships that are not functions.

Identify the repeated input.

x	-1	2	-1
y	3	5	8

7. I can distinguish functions from relationships that are not functions.

For $y = 2x - 5$, find the output when $x = -1$.

8. I can distinguish functions from relationships that are not functions.

The pairs are $(1, 4), (2, 6), (1, 7)$. Is this relation a function? Explain.

I CAN IDENTIFY THE DOMAIN AND RANGE OF A RELATION.

9. I can identify the domain and range of a relation.

Find the range of $\{(1, 4), (2, 4), (3, -1)\}$.

10. I can identify the domain and range of a relation.

The function $f(x) = \sqrt{x}$ has real outputs. Explain why negative inputs are excluded.

11. I can identify the domain and range of a relation.

Tickets cost \$6 each, and at most 8 tickets may be bought. What inputs are reasonable for the total-cost rule?

12. I can identify the domain and range of a relation.

List the domain.

$(2, 5), (4, 7), (6, 5)$

I CAN CONNECT LINEAR GROWTH IN TABLES, GRAPHS, AND EQUATIONS.

13. I can connect linear growth in tables, graphs, and equations.

Write an equation.

Slope 4; y-intercept −2.

14. I can connect linear growth in tables, graphs, and equations.

Find the slope.

Points (1,3) and (5,11)

15. I can connect linear growth in tables, graphs, and equations.

A table has $x = 0, 1, 2, 3$ and $y = 3, 7, 11, 15$. Write a rule.

16. I can connect linear growth in tables, graphs, and equations.

Find the change per step for outputs 5, 9, 13, 17. Is $(1, 2), (2, 2), (1, 3)$ a function?

I CAN CALCULATE AND INTERPRET SLOPE AS A CONSTANT RATE OF CHANGE.

17. I can calculate and interpret slope as a constant rate of change.

Find the slope through $(0, 2)$ and $(3, 8)$.

18. I can calculate and interpret slope as a constant rate of change.

Find the run.

$(-1,8)$ to $(6,10)$

19. I can calculate and interpret slope as a constant rate of change.

As x rises by 4, y falls by 12. Find the slope.

20. I can calculate and interpret slope as a constant rate of change.

One line has slope -2 and another has slope 3. Compare their directions and rates.